

## ESP32 + 2x L298D (4-Motor) Setup Guide

This guide matches the sketch in `2xlu98desp.ino`. Robot layout used: - 2x L298D drivers - 4 DC motors total ## 1) Required Hardware

- ESP32 dev board
- 2x L298D motor driver modules
- 4x DC motors (JGB37 500 RPM in your case)
- 12V battery for motors
- Stable 5V supply for ESP32 (USB or proper regulator)
- Common wiring ground between all devices

### 2) ESP32 to Left L298D Wiring

| ESP32 GPIO | Left L298D Pin | Use                            |
|------------|----------------|--------------------------------|
| 25         | ENA            | PWM speed (left front channel) |
| 32         | IN1            | Direction                      |
| 33         | IN2            | Direction                      |
| 26         | ENB            | PWM speed (left rear channel)  |
| 27         | IN3            | Direction                      |
| 14         | IN4            | Direction                      |

### 3) ESP32 to Right L298D Wiring

| ESP32 GPIO | Right L298D Pin | Use                             |
|------------|-----------------|---------------------------------|
| 18         | ENA             | PWM speed (right front channel) |
| 16         | IN1             | Direction                       |
| 17         | IN2             | Direction                       |
| 19         | ENB             | PWM speed (right rear channel)  |
| 21         | IN3             | Direction                       |
| 22         | IN4             | Direction                       |

### 4) Motor Output Wiring

| Driver      | Output Pins | Motor             |
|-------------|-------------|-------------------|
| Left L298D  | OUT1 + OUT2 | Left Front Motor  |
| Left L298D  | OUT3 + OUT4 | Left Rear Motor   |
| Right L298D | OUT1 + OUT2 | Right Front Motor |
| Right L298D | OUT3 + OUT4 | Right Rear Motor  |

## 5) Power Wiring (Important)

| Source       | Connect To                |
|--------------|---------------------------|
| Battery +12V | +12V on both L298D boards |
| Battery GND  | GND on both L298D boards  |
| ESP32 GND    | Same common ground line   |

Critical rules: - Common ground is mandatory: battery -, both L298D GND, and ESP32 GND must be connected. - Do not power ESP32 from L298D onboard 5V when motors are under load. - Use a separate stable 5V source for ESP32.

## 6) L298D Jumper Note

If your L298D has ENA/ENB jumper caps installed, remove them. Reason: the sketch controls ENA/ENB using PWM from ESP32.

## 7) Control Mapping (Bluepad32)

- Left trigger: forward
- Right trigger: reverse
- Left stick X: steering
- B button: high power mode
- A button: low power mode
- Built-in failsafe: motors stop if input data is lost

## 8) Upload Checklist

1. Open `2xlu98desp.ino` in Arduino IDE.
2. Select the correct ESP32 board and COM port.
3. Verify **Bluepad32** library is installed.
4. Confirm wiring matches the tables above exactly.
5. Upload and pair controller.

Optional pairing reset in sketch: - Set `FORGET_PAISED_KEYS_ON_BOOT = true` once, upload, reboot, pair controller. - Set it back to `false` and upload again.

## 9) First Test Procedure

1. Lift wheels off the ground.
2. Power on ESP32 and motor battery.
3. Pair gamepad.
4. Test low power mode first (A).
5. Check forward/reverse and left/right steering.

If a side runs backward, use these flags in `2x1u98desp.ino`: - `INVERT_LEFT_SIDE`  
- `INVERT_RIGHT_SIDE`

## **10) Performance Note for JGB37 500 RPM**

L298D works, but it has noticeable voltage drop and can heat up with higher current motors.

Expected behavior: - Slightly reduced top speed and torque compared to direct battery voltage - Driver heating during hard acceleration or pushing load

If drivers get too hot, add cooling or move to a higher efficiency driver (for example BTS7960-class).